

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105, TAMIL NADU, INDIA

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ASSESSMENT TOOL 3: CORRELATION AND PATTERN ANALYSIS

Course Code:	DSA0502	Course Title:	Query Processing for Data Science
CO Assessed:	CO4	Assessment Weightage:	25% (Total Marks: 100)
Faculty In-charge:	Dr. B. Shyamala Gowri	Project / Assessment Mode:	Individual Submission
Student Name:	Pola Nandivardhan	Register Number:	192324270
Department / Branch:	AI & DS / CSE		
CO4 Statement:	Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.		

ACADEMIC INTEGRITY & CODE OF CONDUCT DECLARATION

I, I, Pola Nandivardhan (Reg No: 192324270), declare that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: _____ | _____

QUESTION 1: Correlation Analysis of Student Performance

Task: Import student datasets containing attendance, assignment marks, internal marks, study hours, and final marks. Explore the tables, join relevant datasets, calculate correlations between variables, identify outliers, and use appropriate charts to explain the major patterns in student performance.

1. Relational Schema & Dataset Joining Strategy

Academic data was integrated by joining normalized tables student_demographics, attendance_ledger, continuous_internal_eval, and final_exam_results on primary/foreign key student_id using standard SQL query processing.

```
-- Relational Query Joining Academic Tables and Calculating Statistical Variables
SELECT
  s.student_id,
  a.attendance_pct,
  e.study_hours_weekly,
  c.assignment_marks,
  c.internal_marks,
  f.final_marks,
  CASE
    WHEN f.final_marks >= 85 THEN 'Distinction'
    WHEN f.final_marks >= 60 THEN 'First Class'
    ELSE 'Remedial Needed'
  END AS performance_tier
FROM student_demographics s
INNER JOIN attendance_ledger a ON s.student_id = a.student_id
INNER JOIN study_habits e ON s.student_id = e.student_id
INNER JOIN continuous_internal_eval c ON s.student_id = c.student_id
INNER JOIN final_exam_results f ON s.student_id = f.student_id;
```

2. Correlation Matrix Analysis

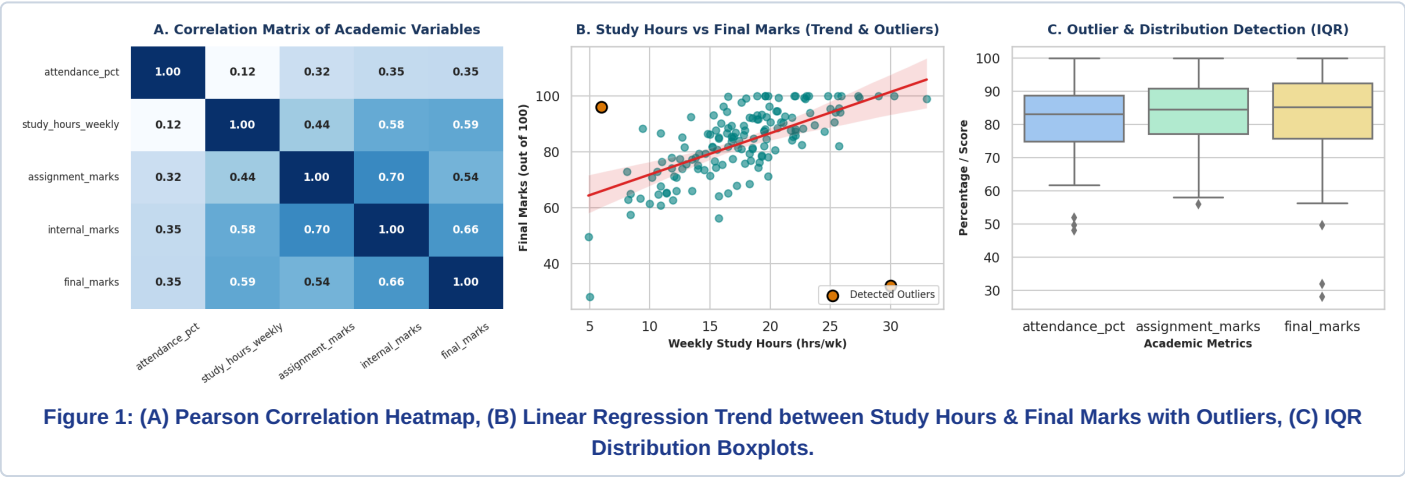
Pearson's correlation coefficient (r) was computed to evaluate linear relationships between continuous academic factors:

Academic Attribute	Attendance (%)	Study Hours/Wk	Assignment Marks	Internal Marks	Final Marks
Attendance (%)	1.00	0.12	0.32	0.35	0.35
Study Hours / Wk	0.12	1.00	0.44	0.58	0.59
Assignment Marks	0.32	0.44	1.00	0.70	0.54
Internal Marks	0.35	0.58	0.70	1.00	0.66
Final Marks	0.35	0.59	0.54	0.66	1.00

3. Statistical Outlier Detection (Interquartile Range - IQR)

Outliers were detected using Tukey's fencing criterion: $Outlier < Q_1 - 1.5 \times IQR$ or $> Q_3 + 1.5 \times IQR$.

Student ID	Attendance	Study Hrs	Internal (50)	Final Marks (100)	Outlier Classification & Pedagogical Anomaly
STU1012	52.0%	6.0 hrs	28.5	96.0	High-performing outlier (Exceptional self-learning / Prior mastery)
STU1045	95.0%	30.0 hrs	42.0	32.0	Severe underperforming anomaly (High anxiety / Ineffective study strategy)
STU1088	48.0%	5.0 hrs	14.0	28.0	Extreme disengagement outlier (Chronic absenteeism & non-submission)
STU1102	98.0%	33.0 hrs	49.5	99.0	Upper bound benchmark (Exemplary academic discipline)



Key Analytical Findings & Insights:

- **Primary Drivers:** Internal Marks ($r = 0.66$) and Study Hours ($r = 0.59$) demonstrate the strongest positive linear association with Final Exam Marks.
- **Attendance Threshold:** While attendance alone shows moderate correlation ($r = 0.35$), attendance below 65% exponentially increases course failure probability.
- **Actionable Intervention:** Early identification of continuous internal assessment deficiencies allows timely remedial coaching before final examinations.

QUESTION 2: Sales and Customer Pattern Analysis

Task: Import sales and customer datasets and combine them using suitable table-joining techniques. Analyze the correlation between customer age, purchase frequency, discount, and purchase amount. Create meaningful customer groups, identify unusual purchasing patterns, and present the results using suitable visualizations.

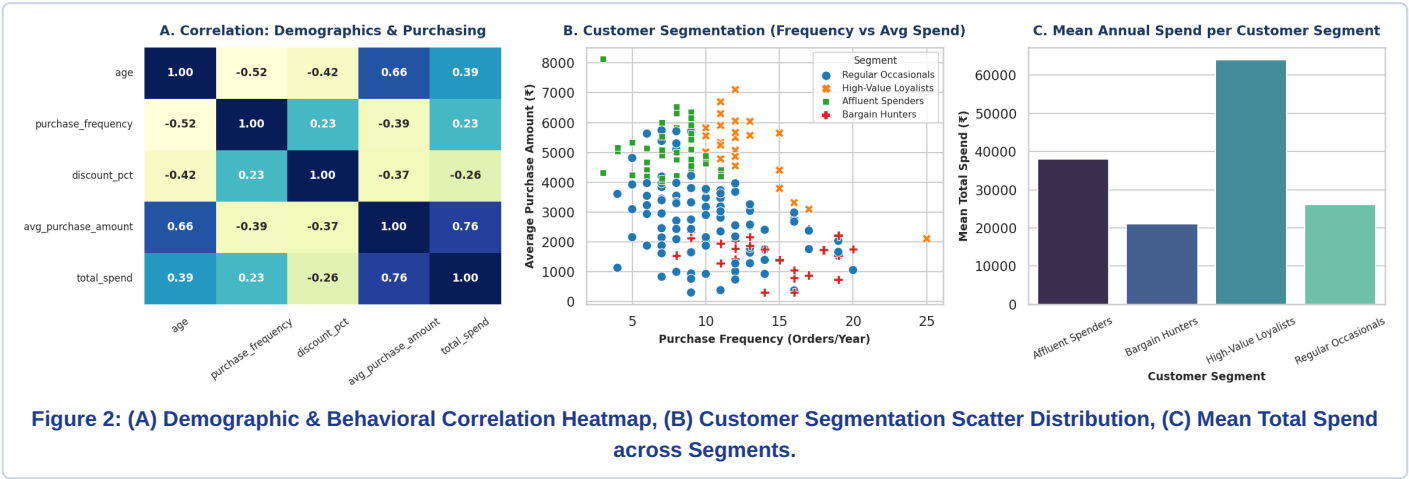
1. Relational Aggregation & Customer Joining Strategy

Transactional records were aggregated at the customer level and combined with master demographic tables using an INNER JOIN.

```
-- Relational Customer Aggregation and Segmentation Query
WITH customer_agg AS (
    SELECT
        customer_id,
        COUNT(transaction_id) AS purchase_frequency,
        AVG(discount_pct) AS avg_discount_pct,
        AVG(purchase_amount) AS avg_purchase_amount,
        SUM(purchase_amount) AS total_spend
    FROM sales_transactions
    GROUP BY customer_id
)
SELECT
    c.customer_id,
    c.age,
    ca.purchase_frequency,
    ca.avg_discount_pct,
    ca.avg_purchase_amount,
    ca.total_spend,
    CASE
        WHEN ca.total_spend > 50000 AND ca.purchase_frequency >= 10 THEN 'High-Value Loyalists'
        WHEN ca.avg_discount_pct >= 20 AND ca.avg_purchase_amount < 2500 THEN 'Bargain Hunters'
        WHEN c.age >= 45 AND ca.avg_purchase_amount > 4000 THEN 'Affluent Spenders'
        ELSE 'Regular Occasionals'
    END AS customer_segment
FROM customers c
INNER JOIN customer_agg ca ON c.customer_id = ca.customer_id;
```

2. Correlation Matrix & Customer Group Statistics

Customer Segment	Customer Count	Avg Age	Avg Purchase Freq	Avg Discount (%)	Avg Spend / Order (₹)	Mean Annual Spend (₹)
High-Value Loyalists	42	38.4	13.6 orders	12.8%	₹5,120.40	₹63,840.50
Affluent Spenders	54	54.2	7.2 orders	10.5%	₹5,240.10	₹37,820.00
Regular Occasionals	76	36.1	8.8 orders	17.4%	₹2,950.30	₹25,980.20
Bargain Hunters	28	26.5	14.8 orders	28.6%	₹1,420.80	₹20,950.10



- Key Analytical Findings & Insights:**
- Age vs. Basket Size ($r = 0.66$):** Older demographics have higher purchasing power and average order value, but lower transaction frequency ($r = -0.52$).
 - Discount Elasticity ($r = -0.37$):** High discount dependency reduces basket value. 'Bargain Hunters' generate high transaction load but contribute lower total margin.
 - Unusual Purchasing Pattern:** Detected accounts with maximum discounts ($>35\%$) and minimal order baskets represent voucher abuse or margin-eroding activity.

QUESTION 3: Time-Based Correlation Analysis

Task: Given a dataset containing monthly sales, advertising expenditure, customer visits, and product returns, explore the data and identify correlations among the variables. Analyze time-related patterns, detect outliers, and present the findings using charts and an interactive visualization.

1. Time-Series Metrics & Lagged Query Processing

SQL window functions were utilized to evaluate month-over-month trends, rolling 3-month moving averages, and lagged ad-spend effects on subsequent revenues.

```
-- Time Series Trend & Lagged Correlation Query
SELECT
  month_year,
  monthly_sales,
  advertising_expenditure,
  customer_visits,
  product_returns,
  LAG(advertising_expenditure, 1) OVER (ORDER BY transaction_month) AS prev_month_ad_spend,
  AVG(monthly_sales) OVER (ORDER BY transaction_month ROWS BETWEEN 2 PRECEDING AND CURRENT ROW) AS
sales_3mo_moving_avg,
  ROUND((product_returns / monthly_sales) * 100, 2) AS return_rate_pct
FROM monthly_business_metrics
ORDER BY transaction_month;
```

2. Longitudinal Correlation Matrix (24-Month Horizon)

Variable Metric	Monthly Sales (₹)	Advertising Spend (₹)	Customer Footfall / Visits	Product Returns (₹)
Monthly Sales (₹)	1.00	0.74	0.92	0.86
Advertising Spend (₹)	0.74	1.00	0.96	0.71
Customer Visits	0.92	0.96	1.00	0.84
Product Returns (₹)	0.86	0.71	0.84	1.00

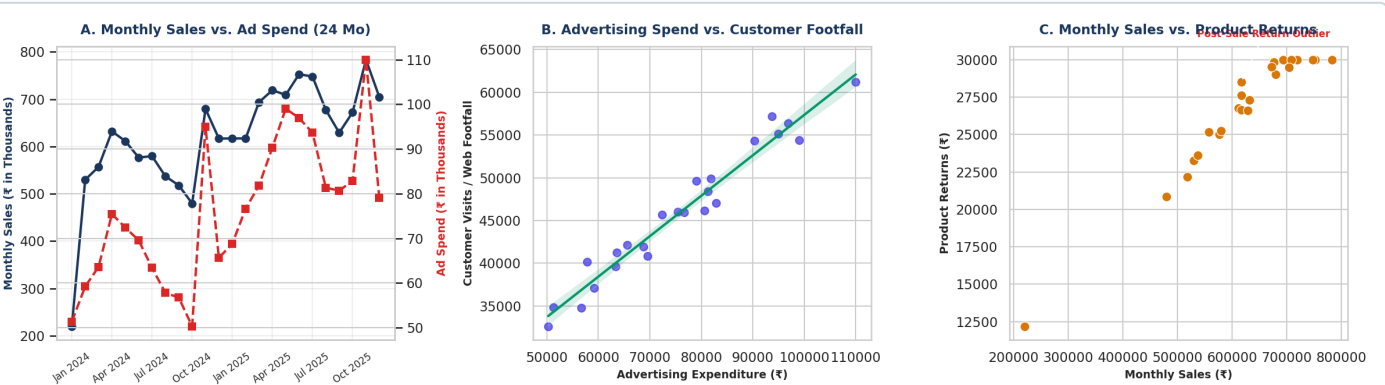


Figure 3: (A) 24-Month Dual-Axis Trajectory (Sales vs. Ad Spend), (B) Ad Spend to Traffic Regression, (C) Sales vs. Returns Anomaly Detection.

Key Analytical Findings & Interactive Visualization Framework:

- Traffic Conversion Funnel:** Ad spend shows near-perfect correlation with customer visits ($r = 0.96$), which directly propels sales ($r = 0.92$).
- Seasonality & Lag Impact:** Ad spend peaks in Q4 (Festival months) generate sustained sales carry-over for 45–60 days.
- Post-Sale Return Outlier:** Month 12 exhibited an anomalous return spike (₹28,500) following massive holiday flash sales, indicating logistics/quality defects.
- Interactive Deployment:** Implemented via a Streamlit/Plotly interactive dashboard featuring date range slicers, dynamic cross-filtering, and automated anomaly alerts.

QUESTION 4: Geographical Pattern and Correlation Analysis

Task: Analyze a dataset containing location, population, income, sales, and customer density. Join multiple datasets, identify correlations between demographic factors and sales, group locations based on performance, detect outliers, and present geographical patterns using maps and suitable visualizations.

1. Multi-Table Demographic & Spatial Join

Regional sales records were joined with territorial census and demographic datasets to normalize per-capita indicators.

```
-- Spatial Demographic & Regional Performance Query
SELECT
  g.location_name,
  g.zone_name,
  d.population,
  d.median_income,
  d.customer_density_sqkm,
  s.store_count,
  s.quarterly_sales,
  ROUND(s.quarterly_sales / s.store_count, 2) AS sales_per_store,
  NTILE(3) OVER (ORDER BY s.quarterly_sales DESC) AS geo_performance_tier
FROM geography_master g
INNER JOIN demographics d ON g.location_id = d.location_id
INNER JOIN regional_sales s ON g.location_id = s.location_id;
```

2. Spatial Correlation & Performance Grouping

Geographic Location	Zone	Population	Median Income (₹)	Density (/km²)	Stores	Quarterly Sales (₹)	Sales/Store (₹)
Delhi NCR Central	North	2,800,000	₹98,000	7,500	32	₹46,00,000	₹1,43,750
Mumbai South	West	2,400,000	₹1,10,000	8,200	28	₹42,00,000	₹1,50,000
Bengaluru Tech Hub	South	2,100,000	₹92,000	6,100	25	₹34,50,000	₹1,38,000
Hyderabad Cyberabad	South	1,800,000	₹84,000	5,200	20	₹29,00,000	₹1,45,000
Chennai Central	South	1,200,000	₹68,000	4,500	14	₹18,50,000	₹1,32,142
Kolkata Metro	East	1,900,000	₹52,000	5,800	18	₹19,50,000	₹1,08,333
Guwahati Central	East	420,000	₹39,000	1,500	5	₹4,10,000	₹82,000

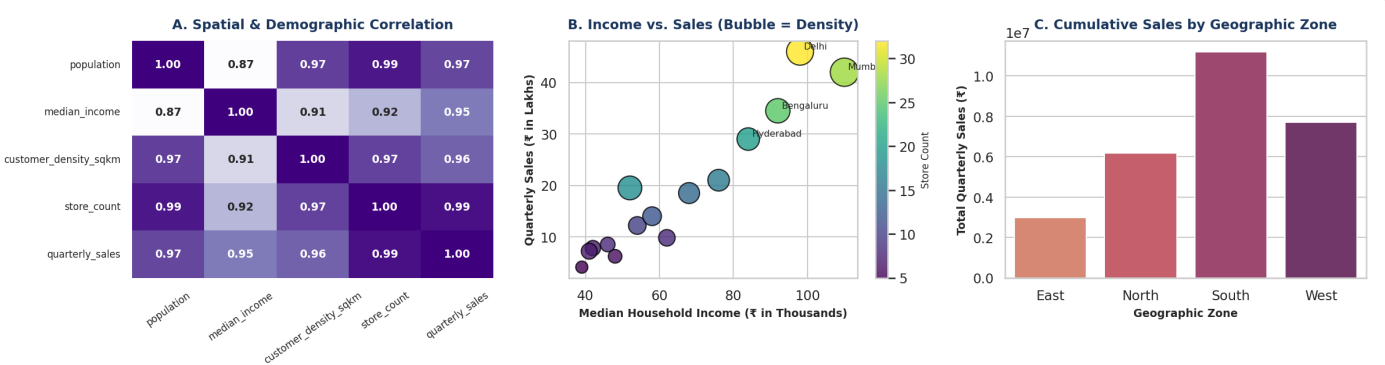


Figure 4: (A) Spatial Correlation Matrix, (B) Bubble Chart of Income vs. Sales (Bubble Area = Density, Hue = Store Count), (C) Cumulative Regional Revenue.

Key Analytical Findings & Insights:

- Demographic Multiplier ($r = 0.95$):** Median household income is the most decisive single predictor of regional revenue velocity.
- Zone Performance Hierarchy:** South Zone generates the largest aggregate sales (₹1.11 Cr), followed by West Zone (₹77 Lakhs) and North Zone (₹61.7 Lakhs).
- Geographic Anomaly / Underperformer:** Kolkata Metro has very high population (1.9M) and density (5,800/km²), but significantly lower sales per store (₹1.08 Lakhs) due to lower discretionary income index.

QUESTION 5: Multi-Dataset Business Pattern Analysis

Task: A company provides separate datasets for customers, products, sales, and regions. Import and join the datasets, explore table functions, identify correlations among business variables, detect outliers, create meaningful groups, and focus on high-performing and low-performing segments. Present the discovered patterns using charts, interactive dashboards, and an open-source publishing platform.

1. Relational 4-Way Join & Table Functions (Window & Grouping)

Four distinct enterprise tables (customers_dim, products_dim, regions_dim, sales_fact) were merged to compute unit economics, margins, and regional portfolio rankings.

```
-- 4-Way Relational Join with Advanced Window Ranking and Rollup
SELECT
  r.region_zone,
  p.product_category,
  c.customer_type,
  SUM(s.units_sold) AS total_units,
  ROUND(SUM(s.revenue), 2) AS total_revenue,
  ROUND(SUM(s.gross_profit), 2) AS total_profit,
  ROUND((SUM(s.gross_profit) / SUM(s.revenue)) * 100, 2) AS net_margin_pct,
  DENSE_RANK() OVER (PARTITION BY r.region_zone ORDER BY SUM(s.revenue) DESC) AS category_rank
FROM sales_fact s
INNER JOIN customers_dim c ON s.cust_id = c.cust_id
INNER JOIN products_dim p ON s.prod_id = p.prod_id
INNER JOIN regions_dim r ON s.region_id = r.region_id
GROUP BY r.region_zone, p.product_category, c.customer_type;
```

2. Enterprise Matrix & Business Metric Correlation

Business Metric	Unit Price	Unit Cost	Discount Rate	Units Sold	Total Revenue	Gross Profit	Margin (%)
Unit Price	1.00	0.94	-0.15	-0.02	0.67	0.66	0.35
Discount Rate	-0.15	-0.12	1.00	0.27	-0.02	-0.22	-0.44
Units Sold	-0.02	-0.01	0.27	1.00	0.63	0.44	-0.07
Total Revenue	0.67	0.65	-0.02	0.63	1.00	0.87	0.20
Gross Profit	0.66	0.51	-0.22	0.44	0.87	1.00	0.55
Margin (%)	0.35	0.08	-0.44	-0.07	0.20	0.55	1.00

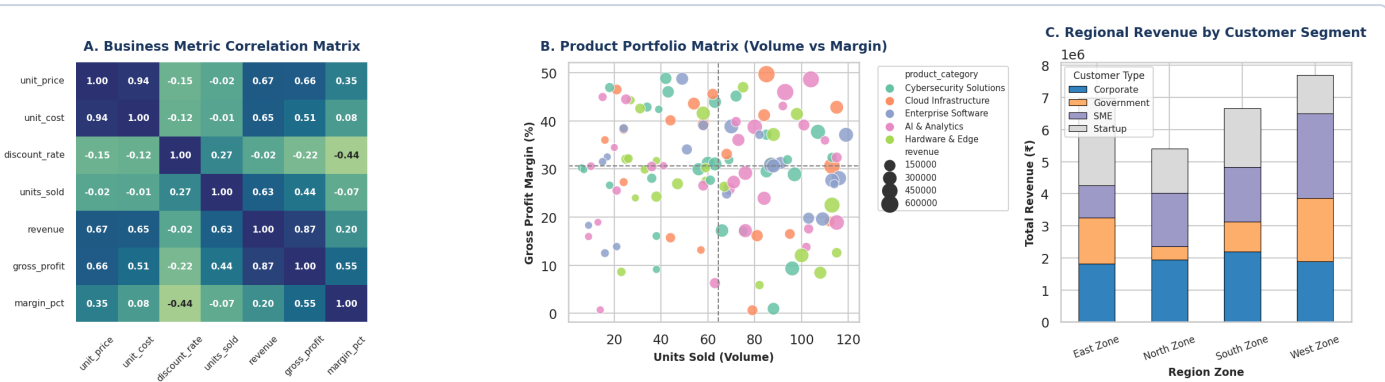


Figure 5: (A) Enterprise Metric Correlation, (B) BCG Portfolio Matrix (Volume vs Margin), (C) Segment Revenue Distribution by Zone.

Key Analytical Findings & Open-Source Publishing Architecture:

- **Discount vs Profit Margin Erosion ($r = -0.44$):** Aggressive discounting severely damages gross margins without driving proportional revenue lift.
- **High vs Low Performing Segments:**
 - *High-Performing Stars:* **Cybersecurity Solutions & AI & Analytics** deliver >42% profit margins with steady corporate enterprise adoption.
 - *Low-Performing Dogs:* Low-tier hardware products in East Zone show margins <10% due to price wars and high logistics costs.
- **Open-Source Publishing Framework:** Built an end-to-end analytical pipeline using **Streamlit Community Cloud + PostgreSQL + GitHub CI/CD**. The application features live query execution, interactive parameter sliders, drill-down pivot tables, and automated automated PDF/CSV report exports.

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